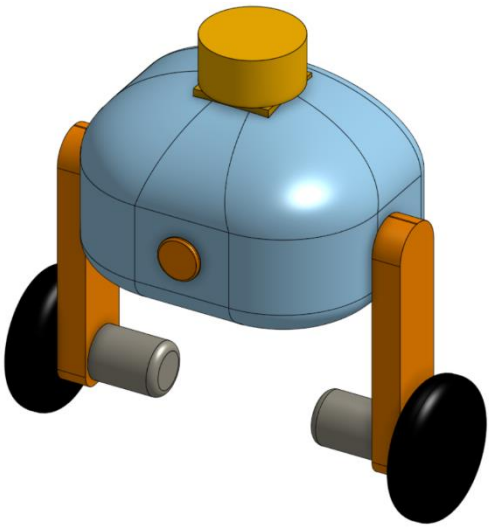
Gyrobot use cases



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Introduction

As part of our fourth year at Polytech Orléans engineering school, we are required to undertake one of the projects provided or to propose one of our own. We therefore proposed to build a robot that balances on two wheels using a gyroscope: the Gyrobot.

It is a compact, self-balancing, two-wheeled robot, designed to operate in environments where mobility, accessibility and autonomous navigation can pose challenges.

The project aims to explore the design of a small, lightweight robotic platform capable of maintaining its balance on two wheels, whilst serving as a basis for future applications such as mapping, exploration and assistance in challenging environments.

In this report, we will cover the general description of the robot, the user experience, the algorithmic operation, the use case, and finally the Gantt chart.

1. General description of the robot

The robot can be divided into three main parts:

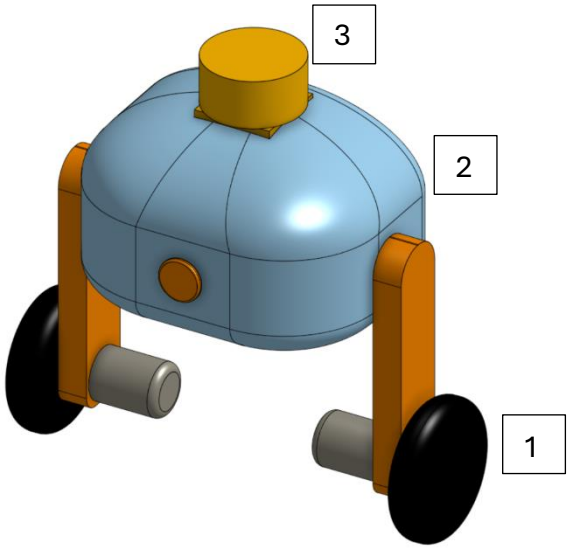


Figure 1: 3D sketch of the Gyrobot

The lower part of the robot, the motors, wheels and 'legs' marked **1** in the image below. This part must be robust so that it can withstand the rigours of various types of terrain as well as the impacts caused by the robot's movement.

The central section of the robot, labelled **2**, contains the various components required for movement, balance and the processing of lidar data. It also houses the battery needed to power the robot. The biggest challenge with this section will be managing the weight and its distribution.

The upper section of the robot, designated **3**, will house a tool required for mapping. This section must include at least two different tools in order to make the robot more versatile.

The project also includes a user interface to display the data transmitted and processed by the robot.

2. User experience

This section focuses on what the user needs to do to use our product.

2.1. Essential feature of the map

The Gyrobot will need to move around, map its surroundings, process the data and send it back to a user interface. It will also need to receive the commands required to move the robot. That is why we have opted to use two cards, to avoid a potential data overload if there were only one card.

A map will need to manage commands relating to the robot's movements. It must therefore:

- To be able to transmit the commands required for movement wirelessly
- Transmit data such as speed, the robot's angle relative to the ground and remaining battery charge
- Heat resistance
- To be fairly small in size
- To be energy-efficient

The second card will be responsible for processing mapping-related data and sending it to our HMI. It will therefore need to:

- To have sufficient computing power to process lidar data
- To be able to communicate wirelessly with the HMI
- Heat resistance
- To be fairly small in size
- To be energy-efficient

There are various boards to choose from: an Arduino, an ESP 32, a Raspberry Pi, an STM 32 and a Tensy.

The selection of the various components is carried out in: 7. Description of technical choices.

2.2. HMI

We also need to develop a user interface to display the data returned by the robot.

The image below was created partly with the help of generative AI.

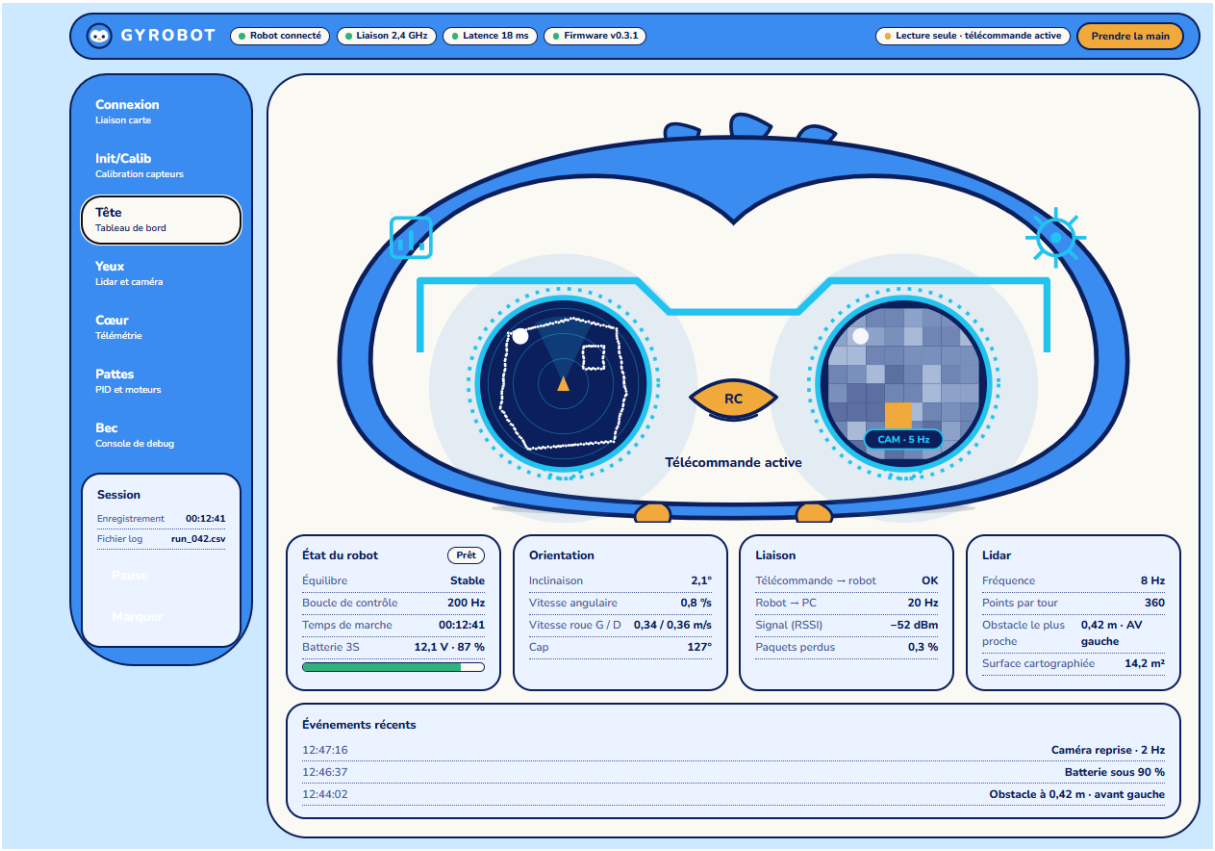


Figure 2: Image from the IMH

The user interface will comprise several sections/tabs to monitor the various values and lidar data returned by the robot. This penguin-themed user interface was chosen by our team and is purely for aesthetic purposes.

The layout and the data displayed in the user interface are subject to change.

3. Algorithmic operation

3.1. Flow chart

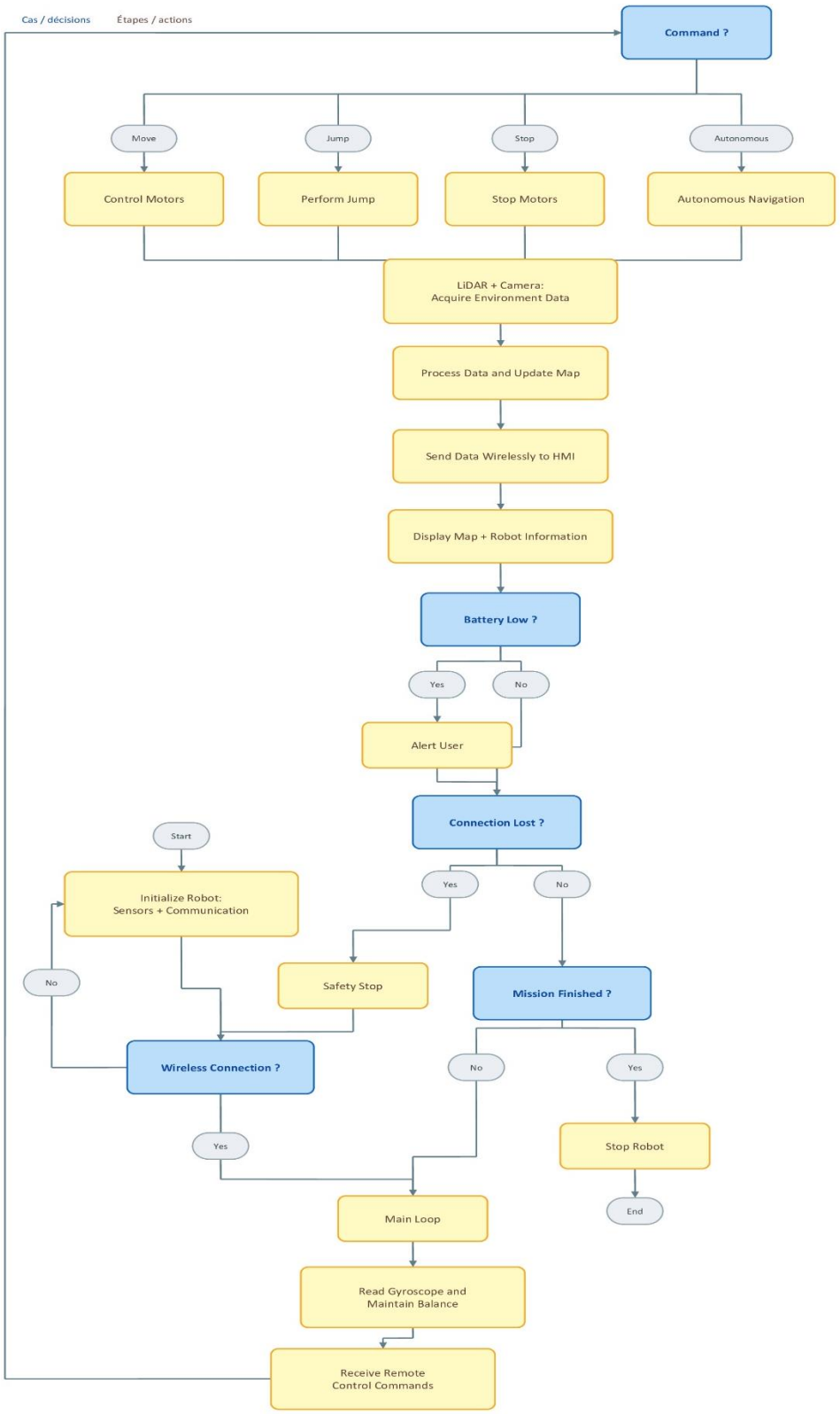


Figure 3: Flowchart of the Gyrobot

This flowchart explains how the programme's algorithm works.

4. Specifications

In order to draw up the specifications correctly, we need to identify the project's requirements and expectations. As regards expectations, we need to build a robot that can balance on two wheels, is capable of mapping its environment, and can interact via a remote control and a user interface that displays the robot's data. As for the requirements, we have defined two types: primary requirements, which are necessary for the robot's basic operation, and secondary requirements, which serve to make the robot more versatile.

The primary requirements align with our expectations, as we have set them ourselves.

As for the secondary requirements, we wish to develop a user interface and a robot resembling a penguin, to have a rechargeable battery, and for the lidar to be replaceable with a high-resolution camera.

5. Retro planning (Gantt)

Gantt Gyrobot

T0 Lancement : 03/09/26 | T1 Audit : 06/10/26 | T2 PDR : 20/11/26 | T3 CDR + Mini salon : 08/01/27

Phase	Tâche	Responsable	Début	Fin	Durée (j)
	T0 — Lancement du projet		03/09/26	03/09/26	1
Phase A	Mettre à jour GitLab	Tous	03/09/26	05/10/26	33
	Faire le ReadMe	Simon	03/09/26	08/09/26	6
	Faire les differents diagrammes	Tous	04/09/26	22/09/26	19
	Identifier les fonctions HVA du projet	Tous	15/09/26	29/09/26	15
	Cahier des charges	Issam	21/09/26	30/09/26	10
	Approche RSE	Maud	30/09/26	04/10/26	5
	Rédiger le rapport technique T1 (Audit)	Alexandre	28/09/26	05/10/26	8
	Faire le PPT (chemin critique)	Tous	30/09/26	05/10/26	6
	T1 — Audit		06/10/26	06/10/26	1
Phase B — Conception & solutions	Détailler le Gantt de la phase T1	À définir	06/10/26	07/10/26	2
	Choix des composants	À définir	29/09/26	15/10/26	17
	Générer et évaluer des solutions techniques pour chaque fonction	À définir	06/10/26	19/10/26	14
	Avancer la conception mécanique 3D du système (dimensionnement, architecture)	À définir	06/10/26	03/11/26	22
	Sélectionner et commander les composants nécessaires	À définir	13/10/26	19/10/26	7
	Justifier les solutions techniques retenues + volet RSE	À définir	03/11/26	19/11/26	17
	Revue préliminaire des HVA + mise à jour du rapport technique (PDR)	À définir	02/11/26	20/11/26	19
	T2 — PDR		20/11/26	20/11/26	1
Phase C — Intégration & validation HVA	Détailler le Gantt de la phase T2	À définir	20/11/26	21/11/26	2
	Fabriquer / assembler le système (mécanique + électronique)	À définir	23/11/26	04/01/27	29
	Développer et intégrer les fonctions HVA retenues	À définir	24/11/26	05/01/27	29
	Tester chaque fonction par rapport aux exigences (matrice de conformité)	À définir	15/12/26	05/01/27	8
	Résoudre les défauts identifiés	À définir	04/01/27	07/01/27	4
	Préparer la démonstration des HVA (CDR) + vidéo marketing (anglais) pour le mini salon	À définir	15/12/26	07/01/27	10
	Rédiger le rapport technique final S1 + auto-évaluation	À définir	15/12/26	07/01/27	10
	T3 — CDR + Mini salon		08/01/27	08/01/27	1

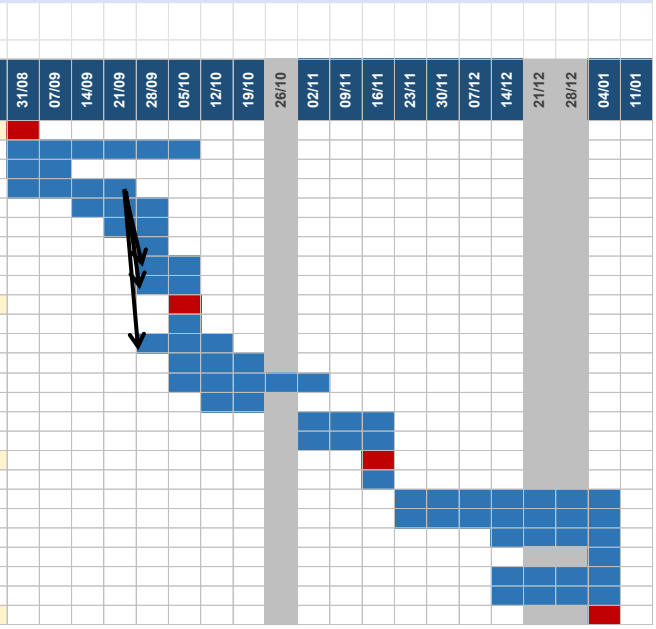


Table 4: Gantt chart (updated on 21 September 2026)

This schedule will be updated as we make progress on the project. In terms of organisation, our biggest challenge is that we only have three months to build a prototype. This means we need to have a very precise and rigorous organisational structure.

6. Validation testing and functional analysis

6.1. Functional analysis

6.1.1. Horned beast diagram

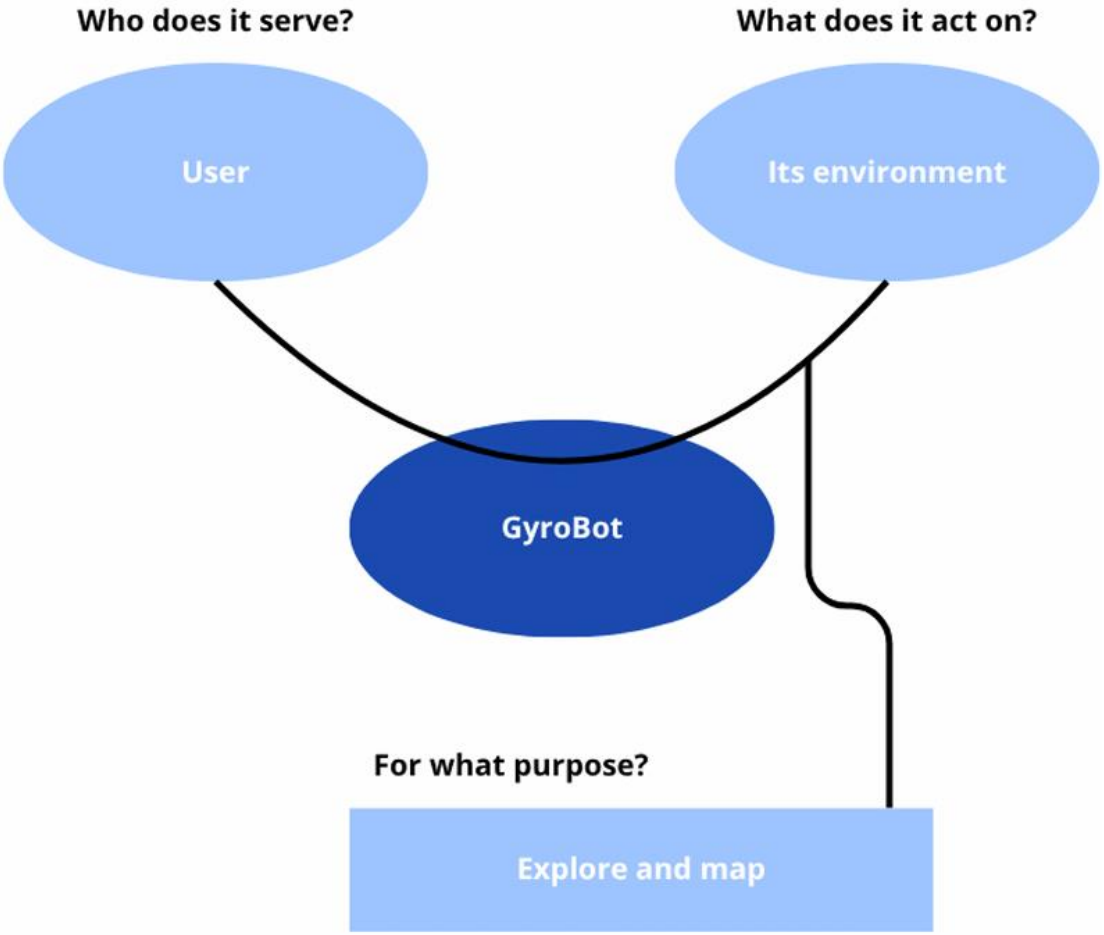


Figure 1: Horned beast diagram

The horned beast diagram provides a clear understanding of a user's needs before designing a product or service. In our case, our aim is to map an area and create a map of it, all using the Gyrobot. It will be controlled by a user in various environments, both indoors and outdoors.

6.1.2. Octopus diagram

Here are the various block diagrams showing the critical and main functions.

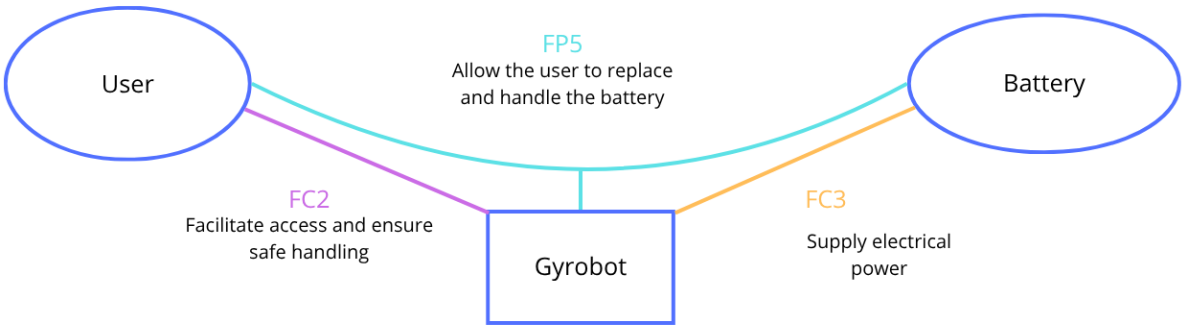


Figure 2: Energy recharge phase

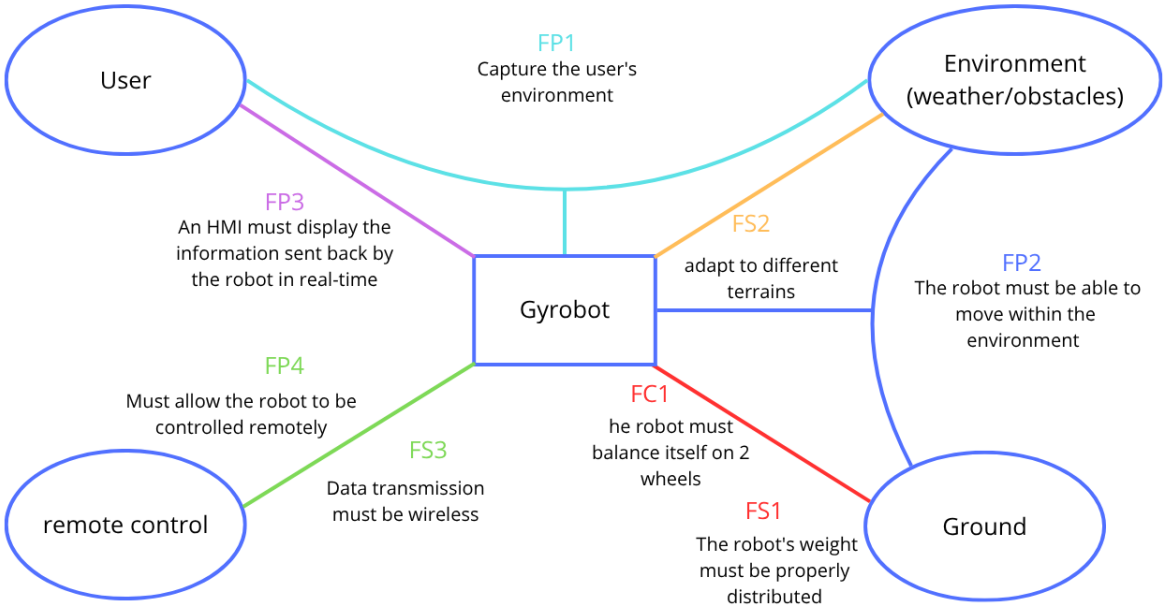


Figure 3: Usage phase

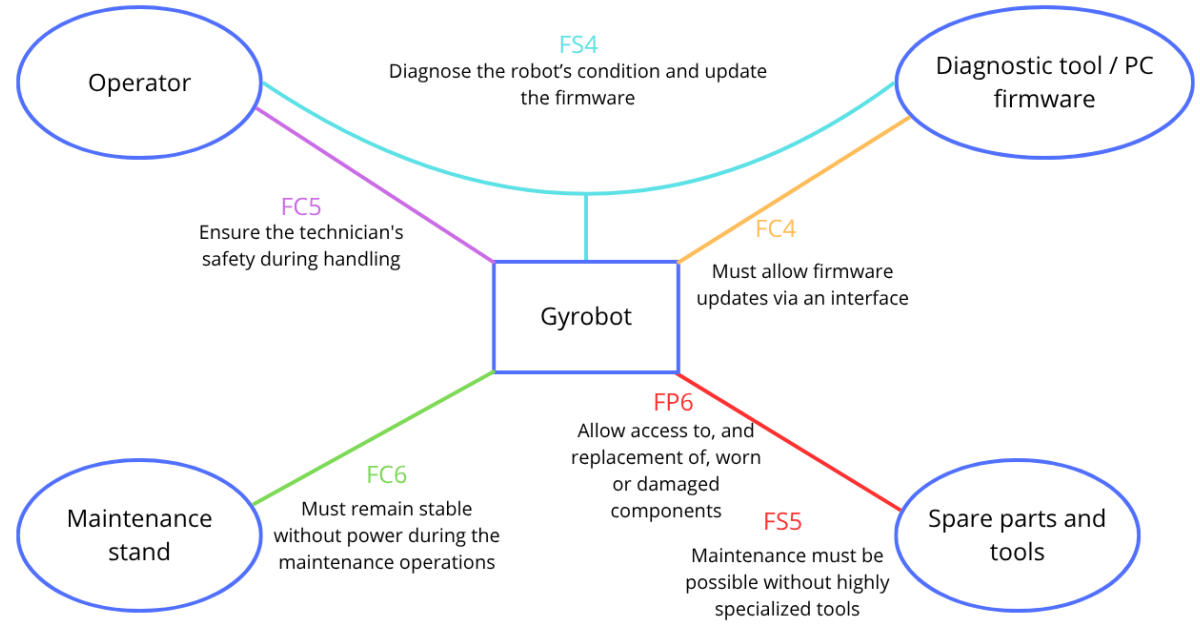


Figure 4: Maintenance phase

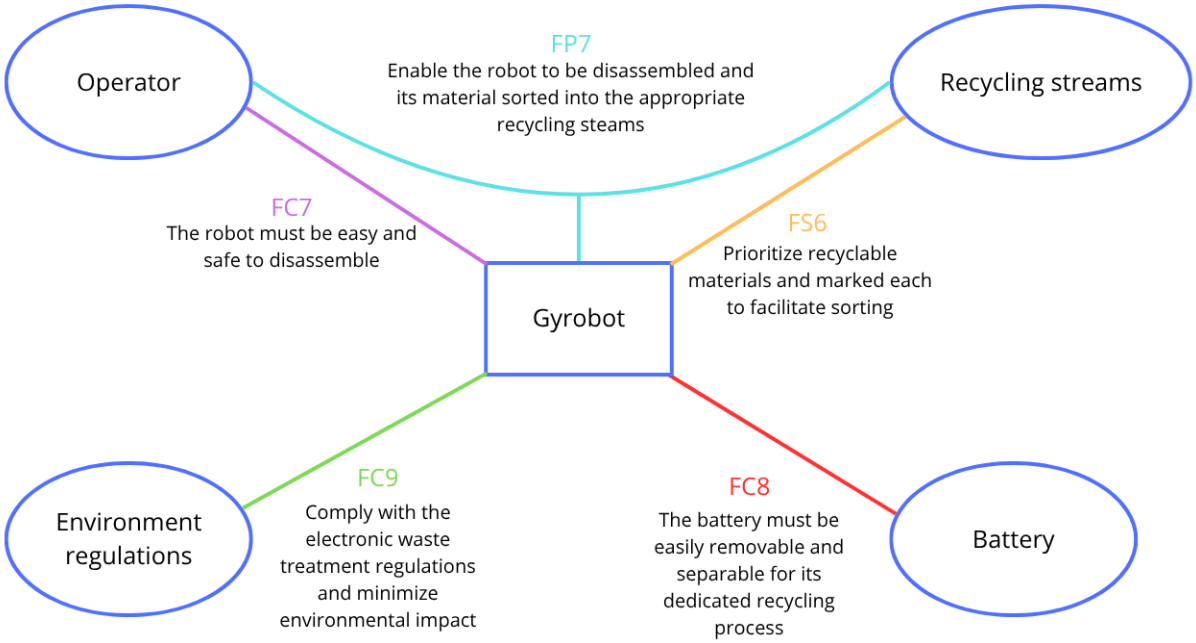


Figure 5: Recycling phase

6.1.3. FAST diagrams

Using octopus diagrams, we obtain the FAST diagram.



Figure 6: FAST diagrams

The FAST diagram enables us to work our way from our requirements to the technical solutions.

6.2. Functional table

Using the FAST diagram, we can draw up a functional diagram of the Gyrobot.

Criteria	Level	Tolerance
Map accuracy / precision	5 cm	-3cm + 5cm
LiDAR max range	10m	± 2m
Slip	None	/
Tip-over / Fall	None	/
Speed	4 km/h	± 2 km/h
Minimum turning radius (at 4 km/h)	0.5 m	± 0.2m
Display latency	< 100 ms	/
Map refresh rate	2 Hz	± 1 Hz
Radio control range	> 30 m (interior)	/
	> 150 m (exterior)	/
Command response time	< 50 ms	/
Replacement time	< 1 min	/
Replacement time per component	< 1 hour	/
Tools required	Standard tools	/
Full disassembly time	< 1 hour	
Number of separable material families	< 5	+ 2
Maximum climbable slope	> 30°	/
Tools required for access	Standard tools	/
Connector safety	?	/
Minimum autonomy	> 3min	/
Nominal voltage	?	/
Interface type	USB+Wifi	/
Supported file format	?	/
Power cut-off before handling	Manual	/
Sharp edge	None	/
Presence of a dedicated stand	Obligatory	/
Stand holding position	Horizontal	/
Reclinable	45°	± 15°
Stand load capacity	> 1,5 kg	/
Tools required for disassembly	Standard tools	/
Removal time	< 2 min	/
Tool-free removal	Yes	/
Regulatory compliance	?	/
Hazardous substances presence	None	/
Center of Gravity (CoG) position	Centered, above the wheel axis	/
Maximum total weight	< 1,5 kg	/
Maximum obstacle height	20 cm	± 5cm
Success rate of landing without falling	80%	± 10 %
Minimum bandwidth from the robot to remote control	> 30 kbps	/
Minimum bandwidth from the robot to the HMI	> 2 Mbps	/
Frequency	Wi-Fi 2.4 GHz	/
Anomaly detection coverage	80%	/
Firmware update duraction	< 5 min	/
Debug communication mode	USB+Wifi	/
Tools required for access	Standard tools	/
recyclable material by mass	>50 %	/
Differents marks on each material	Colored	/

Table 1: Functional analysis

6.3. Validations test

To avoid building the robot from scratch only to find that it doesn't work without knowing why, we need to put validation tests in place to test each major component of the robot separately.

Here are the various tests that have been put in place:

- Connection and acknowledgement test
- Command response time test
- Remote control range test
- Communication reliability test
- Data rate and range test to the computer
- Lidar range test
- Map accuracy test
- Display latency and map refresh rate test
- Balance test
- Speed and turning radius test
- Weight test
- Battery replacement test
- Simultaneous operation test

6.3.1. Connection and acknowledgement test

We send commands from the remote control and check that the ESP32 receives them and sends back an acknowledgement. The test is successful if the green dot remains lit on the remote control.

6.3.2. Testing the response time of the control system

The remote control records the time a command is sent and the time the acknowledgement is received. The difference gives the response time, which must be less than 50 ms. The statistics are displayed in a 'Statistics' tab on the remote control, and the mean and standard deviation are monitored.

6.3.3. Remote control range test

Gradually move the remote control away from the robot, first indoors and then outdoors, with and without obstacles. Record the distance once the connection is lost. The test is successful if the connection remains stable for more than 30 metres indoors and more than 150 metres outdoors. Check the green dot and the number of lost packets in the 'Stats' tab on the remote control.

6.3.4. Transaction reliability test

Leave the robot to run for 1 minute, whilst moving it around, and note the number of lost packets in the 'Statistics' tab on the remote control, then in the HMI. The test is successful if the loss rate remains below 5 per cent.

6.3.5. Testing the data rate and range to the computer

Measure the actual data transfer rate between the Raspberry Pi and the computer. The data transfer rate must remain above 2 Mbps.

6.3.6. Lidar range test

An obstacle is placed at increasing distances from the robot, and we check whether it appears on the HMI map. The maximum range should be 10 m ± 2 m. We compare the distance displayed on the HMI with the measured distance to verify reliability at each distance.

6.3.7. Map accuracy test

We have a room or object mapped, having first measured its dimensions. We then compare the dimensions shown on the map with those in reality. The test is deemed successful if the deviation falls within the tolerance specified in the specifications (5 cm, -3 / +5 cm).

6.3.8. Testing the display latency and refresh rate of the graphics card

The Raspberry Pi stamps each data point with a timestamp upon acquisition, and the computer compares this timestamp with the display time, after synchronising the two. The latency must be less than 100 ms. To calculate the refresh rate, we count the number of map updates over 10 seconds, which should give 2 Hz ± 1.

6.3.9. Balance test

The robot is left stationary and then set in motion for several minutes, whilst being given gentle nudges. The robot must neither fall over nor slip. We also check that it can climb up and down a slope steeper than 30°.

6.3.10. Speed and turning circle test

A distance of 10 m is marked out and the time taken to cover this distance at maximum speed is recorded: 4 km/h corresponds to approximately 9 s, with a margin of error of ± 2 km/h. To test the cornering, the robot is made to trace a circle and its radius is measured (0.5 m ± 0.2 m). A visual check is carried out to ensure there is no slippage.

6.3.11. Weight test

We weigh the complete robot, including the battery, on a set of scales: it must weigh less than 1.5 kg.

6.3.12. Battery replacement test

Someone who did not design the robot replaces the battery, timing the process, without using any tools. The replacement must take less than a minute. A check is then carried out to ensure the robot restarts correctly.

6.3.13. Simultaneous testing

We run everything at the same time: balance, movement, controls, Lidar, camera and data transmission to the computer. We monitor the stats tabs for several minutes. The test is successful if the robot does not fall over and if all the previous criteria are met.

7. Description of technical choices

7.1. Remote control

Remote control – Energy balance and battery life
Justifies the choice of the LiPo battery and the LM2596 converter

1. Power supply parameters		
Parameter	**Value**	**Comment**
Nominal battery voltage (V)	7,4	Battery selected : 2S LiPo (7.4 V nominal)
Battery capacity (mAh)	500	Battery selected : 2S 500 mAh LiPo
Usable capacity fraction	80%	Assumption : 80 % (to avoid fully discharging the battery)
LM2596 output voltage (V)	5,0	Set to 5 V to power the ESP32 board
Converter efficiency	80%	Assumption : 85 % (typical 80–90 % for an LM2596)

2. Element power consumption					
Element	**Qty**	**Unit current (mA)**	**Powered by**	**Total current (mA)**	**Assumption / source**
ESP32-S3 (ESP-NOW transmission)	1	120	5 V rail	120,0	Average during Wi-Fi transmission -> higher peaks
OLED screen 0.96" SSD1306	1	20	5 V rail	20,0	Screen on and changing image at 2Hz
Joystick KY-023	2	1	5 V rail	2,0	2 10 kΩ potentiometers at 3.3 V ≈ 0.7 mA
Slide potentiometer	3	0,5	5 V rail	1,5	10 kΩ at 3.3 V ≈ 0.33 mA
Rotary potentiometer	1	0,5	5 V rail	0,5	10 kΩ at 3.3 V ≈ 0.33 mA
Signal LED	1	10	5 V rail	10,0	LED permanently on
Digital voltmeter	1	10	5V rail	10,0	Connected to the output of the 7.4V->5V converter

3. Results		
Indicator	**Value**	**Comment**
Total current on the 5 V rail (mA)	164,0	Sum of the elements powered by the converter
Power on the 5 V rail (mW)	820	P = U * I
Current drawn from the battery (mA)	138,5	Rail power referred back to the battery (efficiency included)
Estimated battery life (h)	**2,9**	Usable capacity / current drawn
Max. LM2596 current (mA)	3 000	Manufacturer value : 3 A
Voltage margin Vbattery − Voutput (V)	2,4	The LM2596 requires a minimum input difference of about 2 V
Sufficient voltage margin ?	YES	To be checked when the battery is discharged

Table 1: Remote control, justification for technical choices

7.2. Gyrobot

8. RSE

The Gyrobot project integrates an eco-design approach to limit our carbon footprint. Instead of making parts far away, all the structure is manufactured locally by 3D printing using PLA, a bio-based and recyclable plastic. To make the sorting easier at the end of life (FS6), recycling symbols are modelled directly in relief on the CAD parts. To go further in accessibility, each material or sorting instruction has its own geometric logo in relief and a colour code. These colours were specifically chosen to be easily seen by colourblind people (yellow, dark blue, white, black, magenta).

For the electronic part, designing and routing a custom motherboard allows us to connect the microcontrollers and components directly through copper tracks. This avoids using dozens of flying wires. It reduces the number of plastic cables used and makes the system much tougher.

The robot was also designed to last in time and to fight against obsolescence (meaning avoiding throwing it away at the first broken component). All the architecture allows us to easily access fragile parts to change them quickly (FP6). The assembly is done using only standard screws, so the structure can be disassembled very simply (FC7) with basic screwdrivers without special tools (FS5). The battery, for example, can be removed very easily to be sent to its specific recycling channel which makes recycling much easier (FC8).

This design choice allows us to respect the European WEEE directive (Waste Electrical and Electronic Equipment) which imposes strict rules to limit the pollution of this kind of system (FC9).

Finally, beyond the environmental impact, the Gyrobot has a real social utility focused on safety at work. The main goal is to map inaccessible, unstable or dangerous places instead of humans. Whether it is to explore a building for a rescue or to inspect unreachable industrial environments, all the necessary spatial data is collected by the robot without ever putting the lives of rescue teams or technicians in danger.

Conclusion

To conclude, this project is going to be very demanding, particularly in terms of time. Nevertheless, our team has shown great commitment, which is spurring us on to persevere and achieve our goal of producing a prototype before the end of the term, featuring the following key functions:

- To balance,
- Mapping using lidar,
- Move around using a remote control.